

Planning Pseudocode

Note: this is not the pseudocode for the programming assignment. The programming assignment uses a modified version of generic DFS (or CPS's DFS). Part of the assignment is figuring out the appropriate modifications.

STRIPS Algorithm

```
1  def strips( start-list, goal-list)
2      state := start-list
3      plan := []; stack := []
4      push goal-list on stack
5      while stack is not empty
6          strips-step( state, stack, plan)
7      end
8      return plan
9  end
```

STRIPS-step()

```
1  def strips-step( state, stack, plan)
2      next := peek stack
3      if next is a goal satisfied by state
4          pop stack
5      else if next is an unsatisfied conjunctive goal
6          order-goals(next) and push on stack
7      else if next is a single unsatisfied goal
8          choose operator whose postconditions matches goal
9          replace next with operator
10         order-goals(operator preconditions) and push on stack
11     else if next is operator
12         pop stack
13         apply operator postconditions to state
14         add operator to plan
15     end
16 end
```

Discussion

The foundations of the STRIPS algorithm are in DFS and Logic. Because of that, it has an implicit dependency on the unification algorithm.

The main decision points are:

1. order-goals(). Success may depend on the order in which goals are pursued. Therefore the order-goals() may need to keep track of how goals are ordered so that when the algorithm encounters a dead-end, it can re-order the goals and thus enable backtracking.
2. line 8 "choose an operator". Picking the right operator at each step can make a big difference in the operation of the algorithm. What might be a good heuristic for picking operators?
3. line 8 again. Matching postconditions with the goal requires instantiating variables (unification) which variables are instantiated can have an effect on the operation of the algorithm.